

Bahria University, Islamabad

Department of Software Engineering

Machine Learning

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GitHub Link:
https://github.com/Samreen555/Machine-Learning-Course-Assignments-

Notebook Link:
https://colab.research.google.com/drive/1rq_KADOysAZqpRILJ-A-u0EeP0lmV3HL?usp=sharing

Comparative Analysis of Linear Regression and Polynomial Regression for Real Estate Valuation

1. Introduction:

Machine Learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed for every situation. Regression is a supervised machine learning technique used to predict continuous numerical values. It is commonly applied in real estate valuation, sales forecasting, financial analysis, and price prediction.

Linear Regression is one of the simplest regression techniques. It models the relationship between input and output variables using a straight line. However, some real-world relationships are nonlinear, meaning that a straight line may not adequately represent the relationship between the variables. Polynomial Regression addresses this limitation by introducing higher powers of the input features, such as x^2, x^3, and x^4. These additional features allow the model to represent curved relationships while continuing to use Linear Regression to estimate the coefficients.

In this assignment, Linear Regression and Polynomial Regression models are implemented using a real estate valuation dataset. The models are evaluated and compared to investigate the effect of polynomial degree on predictive performance, model complexity, underfitting, and overfitting.

2. Problem Statement

Real estate prices can vary according to several factors, including property age, location, transportation accessibility, and nearby facilities.

The objective of this assignment is to investigate whether Polynomial Regression can predict house price per unit area more effectively than Linear Regression using the distance to the nearest Mass Rapid Transit (MRT) station as the input feature. Different polynomial degrees are evaluated to determine how model complexity affects prediction accuracy and generalization to unseen observations.

3. Objectives

The main objectives of this assignment are:

1. Load and inspect a real-world regression dataset.
2. Identify an appropriate numerical input feature and continuous target variable.
3. Divide the dataset into training and testing sets.
4. Implement a Linear Regression baseline model.
5. Train Polynomial Regression models using different polynomial degrees.
6. Evaluate model performance using R^2, Mean Squared Error, Root Mean Squared Error and Mean Absolute Error.
7. Apply five-fold cross-validation to compare the polynomial models.
8. Investigate underfitting and overfitting using training and testing performance.
9. Predict a new house price using the trained models.
10. Select a suitable polynomial degree based on experimental results.

4. Dataset Description

4.1 Dataset Name

Real Estate Valuation Dataset

4.2 Dataset Overview

The dataset contains real estate observations with information about transaction dates, house age, distance to the nearest MRT station, number of convenience stores, geographic coordinates, and house price per unit area.
The dataset was provided in Microsoft Excel format and loaded into Python using the Pandas library.

4.3 Dataset Characteristics

The dataset inspection produced the following results:

Characteristic	Result
Total observations	414
Total columns	8
Missing values	0
Duplicate rows	0
Input feature selected	Distance to the nearest MRT station
Target variable	House price per unit area
Machine learning task	Regression

4.4 Description of Dataset Columns

Column	Description
No	Record identifier
X1 transaction date	Date associated with the transaction
X2 house age	Age of the house
X3 distance to the nearest MRT station	Distance between the property and the nearest MRT station
X4 number of convenience stores	Number of nearby convenience stores
X5 latitude	Geographic latitude of the property
X6 longitude	Geographic longitude of the property
Y house price of unit area	House price per unit area

4.5 Selected Input and Target Variables

For this experiment, one numerical input feature was selected.

Input variable (X): Distance to the nearest MRT station.

Target variable (y): House price per unit area.

The record identifier and other property features were excluded from this experiment. This allows the relationship between one numerical input feature and the target variable to be studied and visualized more clearly.

4.6 Statistical Analysis

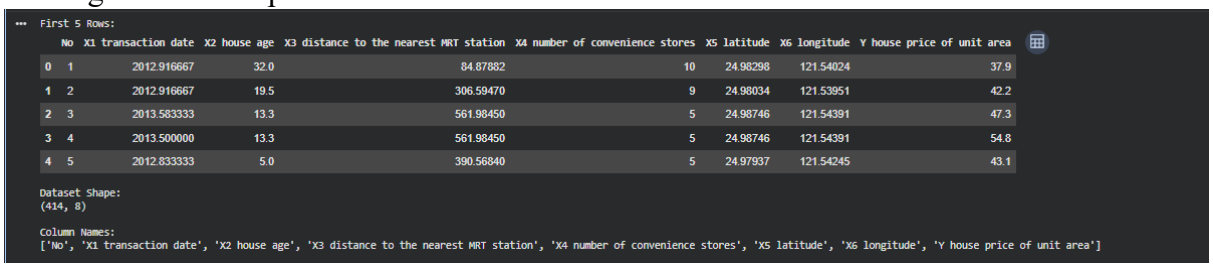
The descriptive statistics generated from the dataset are summarized below.

Statistic	Distance to Nearest MRT Station	House Price per Unit Area
Count	414	414
Mean	1083.8857	37.9802
Standard deviation	1262.1096	13.6065
Minimum	23.3828	7.6
25th percentile	289.3248	27.7
Median	492.2313	38.45
75th percentile	1454.2790	46.6
Maximum	6488.0210	117.5

The mean house price per unit area is approximately 37.98, while the standard deviation is approximately 13.61. The observed prices range from 7.6 to 117.5 in the dataset's reported units.
The distance to the nearest MRT station ranges from approximately 23.38 to 6488.02. This indicates considerable variation in the properties' proximity to public transportation.

4.7 Data Quality Inspection

The dataset was inspected using Pandas functions, including head(), shape, info(), isnull().sum(), describe(), and duplicated().sum().
The inspection confirmed that all 414 observations contained values in every column. No missing values or duplicate rows were detected.



First 5 Rows:

	No	X1 transaction date	X2 house age	X3 distance to the nearest MRT station	X4 number of convenience stores	X5 latitude	X6 longitude	Y house price of unit area
0	1	2012.916667	32.0	84.87882	10	24.98298	121.54024	37.9
1	2	2012.916667	19.5	306.59470	9	24.98034	121.53951	42.2
2	3	2013.583333	13.3	561.98450	5	24.98746	121.54391	47.3
3	4	2013.500000	13.3	561.98450	5	24.98746	121.54391	54.8
4	5	2012.833333	5.0	390.56840	5	24.97937	121.54245	43.1

Dataset Shape:
(414, 8)

Column Names:
['No', 'X1 transaction date', 'X2 house age', 'X3 distance to the nearest MRT station', 'X4 number of convenience stores', 'X5 latitude', 'X6 longitude', 'Y house price of unit area']

Screenshot 1: First five rows and dataset shape

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 414 entries, 0 to 413
Data columns (total 8 columns):
 #   Column                                  Non-Null Count  Dtype
---  ------                                  --------------  -----
 0   No                                      414 non-null    int64
 1   X1 transaction date                     414 non-null    float64
 2   X2 house age                            414 non-null    float64
 3   X3 distance to the nearest MRT station  414 non-null    float64
 4   X4 number of convenience stores         414 non-null    int64
 5   X5 latitude                             414 non-null    float64
 6   X6 longitude                            414 non-null    float64
 7   Y house price of unit area              414 non-null    float64
dtypes: float64(6), int64(2)
memory usage: 26.0 KB

Missing Values:
No                                      0
X1 transaction date                     0
X2 house age                            0
X3 distance to the nearest MRT station  0
X4 number of convenience stores         0
X5 latitude                             0
X6 longitude                            0
Y house price of unit area              0
dtype: int64

Statistical Summary:
```

	No	X1 transaction date	X2 house age	X3 distance to the nearest MRT station	X4 number of convenience stores	X5 latitude	X6 longitude	Y house price of unit area
count	414.000000	414.000000	414.000000	414.000000	414.000000	414.000000	414.000000	414.000000
mean	207.500000	2013.148953	17.712560	1083.885689	4.094203	24.969030	121.533361	37.980193
std	119.655756	0.281995	11.392485	1262.109595	2.945562	0.012410	0.015347	13.606488
min	1.000000	2012.666667	0.000000	23.382840	0.000000	24.932070	121.473530	7.600000
25%	104.250000	2012.916667	9.025000	289.324800	1.000000	24.963000	121.528085	27.700000
50%	207.500000	2013.166667	16.100000	492.231300	4.000000	24.971100	121.538630	38.450000
75%	310.750000	2013.416667	28.150000	1454.279000	6.000000	24.977455	121.543305	46.600000
max	414.000000	2013.583333	43.800000	6488.021000	10.000000	25.014590	121.566270	117.500000

```
Duplicate Rows:
0
```

Screenshot 2: Dataset information, missing values, and statistical summary

5. Tools and Technologies Used

The following tools and libraries were used to implement the experiment.

Tool or Library	Purpose
Google Colab	Cloud-based Python development environment
Python	Programming language
Pandas	Dataset loading and data analysis
NumPy	Numerical calculations
Matplotlib	Data visualization
Scikit-learn	Regression models, preprocessing, evaluation, and cross-validation
Microsoft Excel	Original dataset format

6. Theoretical Background

6.1 Linear Regression

Linear Regression is a supervised learning algorithm that models the relationship between an input variable and a continuous target using a straight-line equation.
For one input feature, the general equation is:

$$y = \beta_0 + \beta_1 x$$

Where:

- y represents the predicted target value.
- x represents the input feature.
- β_0 represents the intercept.
- β_1 represents the coefficient of the input feature.

Linear Regression was used as the baseline model in this assignment.
A baseline provides a reference point for determining whether the more complex Polynomial Regression models improve predictive performance.

6.2 Polynomial Regression

Polynomial Regression extends Linear Regression by introducing higher powers of the input feature.
Its general equation is:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \ldots + \beta_n x^n$$

Where:

- x is the original input feature.
- x^2, x^3, and higher powers are polynomial features.
- β_0, β_1, β_2, and subsequent coefficients are learned by the model.
- n represents the polynomial degree.

Polynomial Regression can represent curved relationships that a straight-line model may fail to capture.

6.3 Polynomial Degrees Used

The following polynomial degrees were evaluated:

Degree	Features Generated	Model Flexibility
Degree 1	x	Linear relationship
Degree 2	x, x^2	Quadratic relationship
Degree 3	x, x^2, x^3	Cubic relationship
Degree 4	x, x^2, x^3, x^4	More flexible polynomial relationship
Degree 5	x, x^2, x^3, x^4, x^5	Greater flexibility

As the polynomial degree increases, the model can represent more complex relationships. However, higher degrees may also increase the risk of overfitting.
Degree 1 Polynomial Regression is mathematically equivalent to Linear Regression when both use the same single input feature.

6.4 Underfitting

Underfitting occurs when a model is too simple to capture important patterns in the data. It is commonly associated with poor performance on both training and testing data.
For example, a straight line may underfit data if the underlying relationship is substantially curved.

6.5 Overfitting

Overfitting occurs when a model learns patterns specific to the training data that do not generalize well to unseen observations. A typical sign is very good training performance combined with noticeably worse testing performance.
Higher polynomial degrees can increase model flexibility and may lead to overfitting if the model begins capturing noise rather than meaningful patterns.

6.6 Generalization

Generalization refers to a model's ability to make accurate predictions on data that was not used during training.
A model should not be selected solely because it fits the training data well. Its performance on unseen data and its cross-validation results should also be considered.

7. Methodology

The experiment followed a structured machine learning workflow.

7.1 Dataset Loading

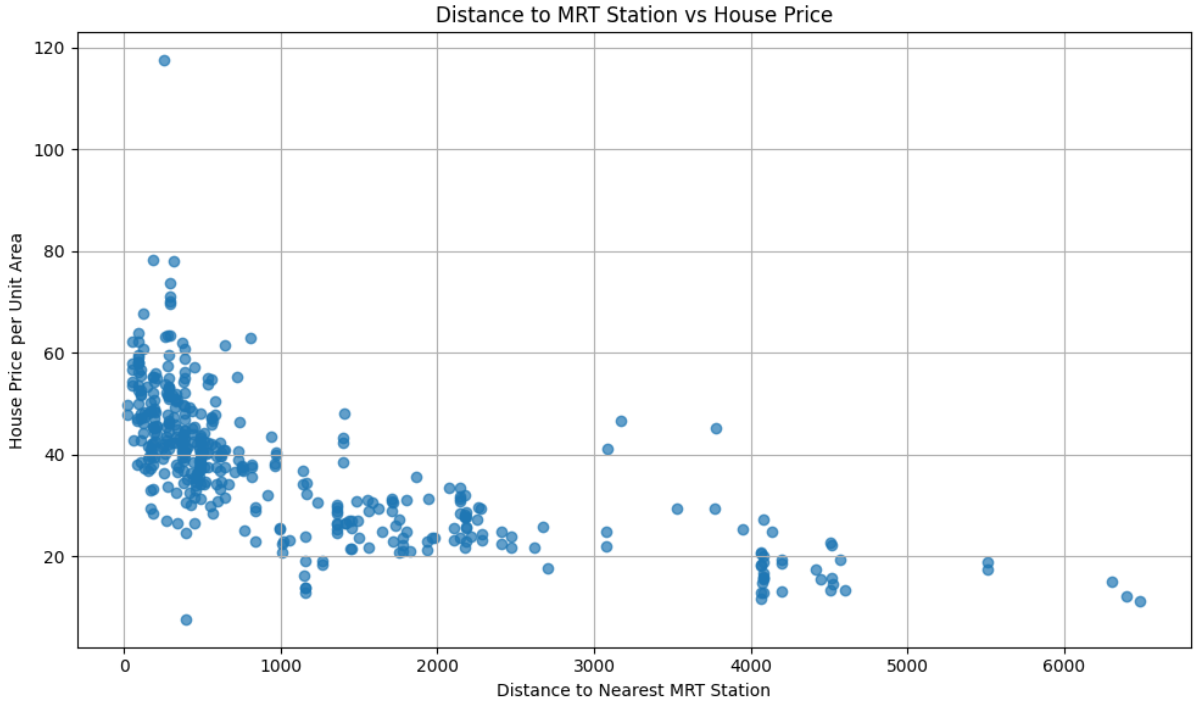
The Excel dataset was loaded into a Pandas DataFrame using pd.read_excel(). The dataset was inspected to identify its dimensions, columns, data types, missing values, duplicate records, and statistical characteristics.

7.2 Feature Selection

The distance to the nearest MRT station was selected as the input variable. House price per unit area was selected as the target variable. Only one input feature was used to make the relationship easier to visualize and interpret.

7.3 Exploratory Data Visualization

A scatter plot was generated to examine the relationship between distance to the nearest MRT station and house price per unit area. The visualization was used to inspect the distribution of observations and assess whether a straight-line relationship appeared sufficient.



Screenshot 3: Scatter plot of distance to MRT station versus house price

7.4 Train-Test Split

The dataset was divided into training and testing sets using Scikit-learn's train_test_split() function.
The split configuration was:

- Training proportion: 80%
- Testing proportion: 20%
- Random state: 27

For 414 observations, this produced 331 training observations and 83 testing observations. The training set was used to fit the models, while the test set was reserved for evaluating performance on unseen observations.

7.5 Model Training

- A Linear Regression model was trained as the baseline.
- Polynomial Regression models of Degrees 1, 2, 3, 4, and 5 were then trained using polynomial feature generation followed by Linear Regression.
- A Scikit-learn pipeline was used to organize preprocessing and model training.
- StandardScaler was included to standardize the input feature. Polynomial features were generated within each model pipeline.
- This approach ensures that the preprocessing transformations are fitted using the relevant training data rather than learning their parameters from the entire dataset before evaluation.

7.6 Model Evaluation

The trained models were evaluated using the same training and testing split.
The following metrics were calculated:

- R^2 Score
- Mean Squared Error
- Root Mean Squared Error
- Mean Absolute Error

Training and testing R^2 scores were also compared to investigate potential underfitting and overfitting.

7.7 Cross-Validation

- Five-fold cross-validation was used to evaluate the polynomial models across multiple data splits.
- The data was divided into five folds. During each iteration, four folds were used for training and one fold was used for validation.
- The process was repeated until every fold had been used as the validation set.
- The mean cross-validation R^2 score was used to compare average performance, while the standard deviation was examined to assess variation across the folds.

7.8 Prediction of a New Value

- A new distance value was supplied to the trained models. Each model generated a predicted house price per unit area.
- The predictions were compared to investigate how model complexity affected the predicted result.
- Predictions within the observed input range were preferred because polynomial models may behave unreliably when extrapolating beyond the range of the training data.

8. Evaluation Metrics

8.1 R^2 Score

R^2 measures how much of the variation in the target variable is explained by the model relative to a baseline that predicts the mean target value.
The general formula is:

$$R^2 = 1 - (SS_{res} / SS_{tot})$$

Where:

- SS_{res} is the sum of squared residuals.
- SS_{tot} is the total sum of squares.

A higher R^2 is generally preferable. A negative R^2 is possible when a model performs worse than predicting the mean of the target values in the test set.

8.2 Mean Squared Error

Mean Squared Error measures the average squared difference between actual and predicted values.

$$MSE = (1/n) \Sigma(y_i - \hat{y}_i)^2$$

Where:

- y_i represents the actual value.
- $\hat{y}_i$ represents the predicted value.
- n represents the number of observations.

A lower MSE indicates smaller squared prediction errors.

8.3 Root Mean Squared Error

Root Mean Squared Error is calculated as:

$$RMSE = \sqrt{MSE}$$

RMSE is expressed in the same units as the target variable and provides a measure of the typical scale of prediction error, with greater weight given to larger errors.
A lower RMSE is generally preferable.

8.4 Mean Absolute Error

Mean Absolute Error measures the average absolute difference between actual and predicted values.

$$MAE = (1/n) \Sigma|y_i - \hat{y}_i|$$

A lower MAE indicates a smaller average absolute prediction error.

9. Experimental Results

9.1 Linear Regression Baseline

The Linear Regression model was trained using the training dataset and evaluated on the test dataset. The baseline model provides a reference for assessing whether polynomial feature expansion improves predictive performance.

Baseline results:

- Training R^2: 0.4317
- Test R^2: 0.5390
- Test MSE: 77.3440
- Test RMSE: 8.7945
- Test MAE: 6.9210

9.2 Comparison of Polynomial Regression Models

Polynomial Regression models of Degrees 1, 2, 3, 4, and 5 were trained and evaluated.

	Model	Training R2	Test R2	Test MSE	Test RMSE	Test MAE
0	Linear Regression	0.4317	0.5390	77.3440	8.7945	6.9210
1	Polynomial Degree 1	0.4317	0.5390	77.3440	8.7945	6.9210
2	Polynomial Degree 2	0.5074	0.5865	69.3633	8.3285	6.1060
3	Polynomial Degree 3	0.5343	0.6352	61.1931	7.8226	5.7591
4	Polynomial Degree 4	0.5465	0.6443	59.6755	7.7250	5.5760
5	Polynomial Degree 5	0.5477	0.6281	62.3907	7.8988	5.7325

Results Interpretation:

The Linear Regression model achieved a test R^2 of **0.5390**. Polynomial Regression Degree 4 achieved the highest test R^2 of **0.6443**, while Degree 4 also achieved the lowest test RMSE of **7.7250**. These results indicate that polynomial feature expansion **did improve performance compared with the baseline — the best model (Degree 4) improved R^2 by approximately 0.105 and reduced RMSE by approximately 1.07 units**. The model with the highest polynomial degree (Degree 5) was not the best performer, confirming that performance was evaluated using multiple metrics and cross-validation rather than assumed.

Model Performance Comparison:

	Model	Training R2	Test R2	Test MSE	Test RMSE	Test MAE
0	Linear Regression	0.4496	0.4534	91.8034	9.5814	7.4302
1	Polynomial Degree 1	0.4496	0.4534	91.8034	9.5814	7.4302
2	Polynomial Degree 2	0.5233	0.5268	79.4703	8.9146	6.7304
3	Polynomial Degree 3	0.5588	0.5281	79.2619	8.9029	6.8516
4	Polynomial Degree 4	0.5758	0.5105	82.2183	9.0674	7.0111
5	Polynomial Degree 5	0.5763	0.5093	82.4225	9.0787	7.0096

Screenshot 4: Model performance comparison table

9.3 Training Versus Testing Performance

The training and testing scores were compared to investigate how well the models generalized to unseen observations.
The R^2 gap is calculated as:

$$\textbf{R}^2 \textbf{ Gap} = \textbf{Training } R^2 - \textbf{Test } R^2$$

A substantial positive gap may indicate that the model performs better on the training data than on unseen data.
However, the gap must be interpreted alongside the absolute training and testing scores. A small gap alone does not prove good model performance.

Training vs Testing R2 Comparison:

	Degree	Training R2	Test R2	R2 Gap	Test RMSE
1	1	0.4496	0.4534	-0.0038	9.5814
2	2	0.5233	0.5268	-0.0035	8.9146
3	3	0.5588	0.5281	0.0307	8.9029
4	4	0.5758	0.5105	0.0653	9.0674
5	5	0.5763	0.5093	0.0670	9.0787

Screenshot 5: Training and testing comparison

9.4 Cross-Validation Results

Five-fold cross-validation was used to compare the performance of the polynomial models across different data splits.

	Degree	Fold 1 R2	Fold 2 R2	Fold 3 R2	Fold 4 R2	Fold 5 R2	Mean CV R2	Std CV R2
0	1	0.5390	0.3211	0.5449	0.4680	0.3601	0.4466	0.0915
1	2	0.5865	0.3603	0.6478	0.5389	0.4291	0.5125	0.1045
2	3	0.6352	0.4072	0.6900	0.5325	0.4580	0.5446	0.1057
3	4	0.6443	0.4205	0.7129	0.5289	0.4737	0.5561	0.1080
4	5	0.6281	0.4203	0.7083	0.5251	0.4743	0.5512	0.1042

Best Degree Based on Mean Cross-Validation R2: 4
Mean Cross-Validation R2: 0.5561

The highest mean cross-validation R^2 was achieved by **Degree 4**, with a mean score of **0.5561**. This suggests that **Degree 4** achieved the strongest average validation performance among the tested polynomial models. The standard deviation of the cross-validation scores was **0.1080**. This value helps describe how much the model's validation performance varied across the five folds. Cross-validation provides additional evidence for model selection, although its estimates may still vary because the dataset contains a limited number of observations.

```
Five-Fold Cross-Validation Results:
```

	Degree	Fold 1 R2	Fold 2 R2	Fold 3 R2	Fold 4 R2	Fold 5 R2	Mean CV R2	Std CV R2
0	1	0.5390	0.3211	0.5449	0.4680	0.3601	0.4466	0.0915
1	2	0.5865	0.3603	0.6478	0.5389	0.4291	0.5125	0.1045
2	3	0.6352	0.4072	0.6900	0.5325	0.4580	0.5446	0.1057
3	4	0.6443	0.4205	0.7129	0.5289	0.4737	0.5561	0.1080
4	5	0.6281	0.4203	0.7083	0.5251	0.4743	0.5512	0.1042

```
Best Degree Based on Mean Cross-Validation R2:
4
Mean Cross-Validation R2: 0.5561
```

Screenshot 6: Cross-validation results table

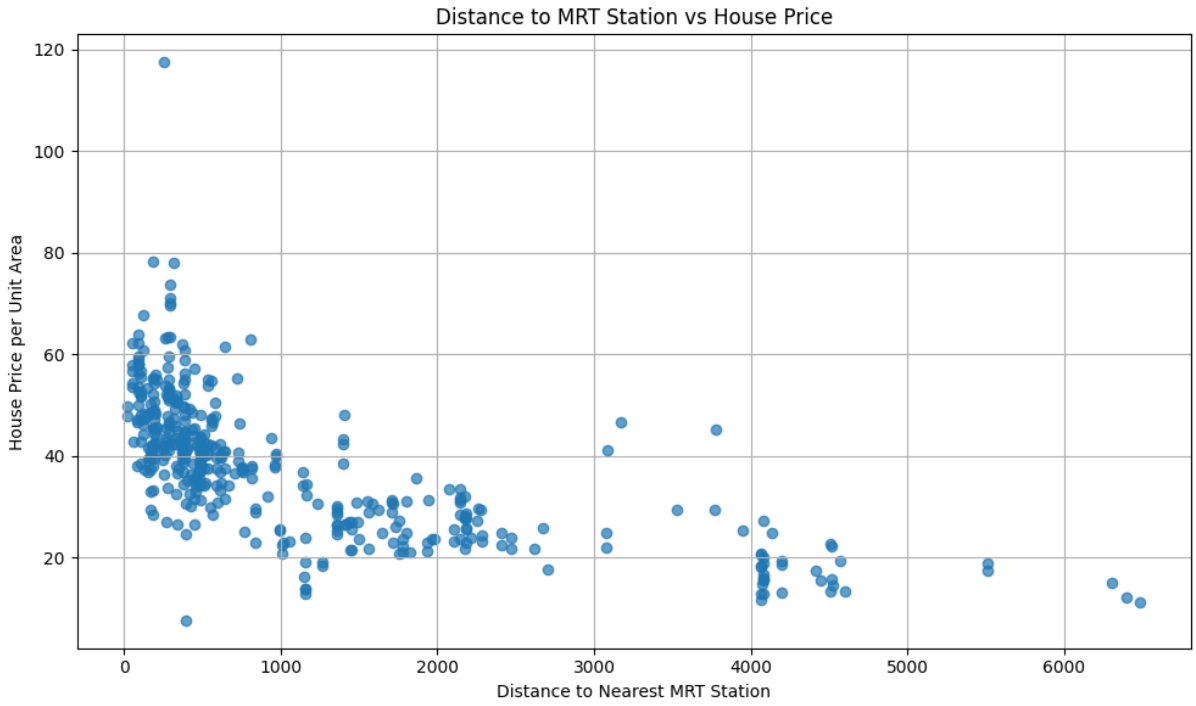
```
Best Polynomial Degree: 4
Mean Cross-Validation R2: 0.5561

Polynomial Equation
Note: z is the standardized distance to the nearest MRT station.
Predicted House Price = 30.188518 - 12.712756 * z^1 + 14.227865 * z^2 - 6.614109 * z^3 + 0.889884 * z^4
```

10. Visualization and Graphical Analysis

10.1 Original Dataset Scatter Plot

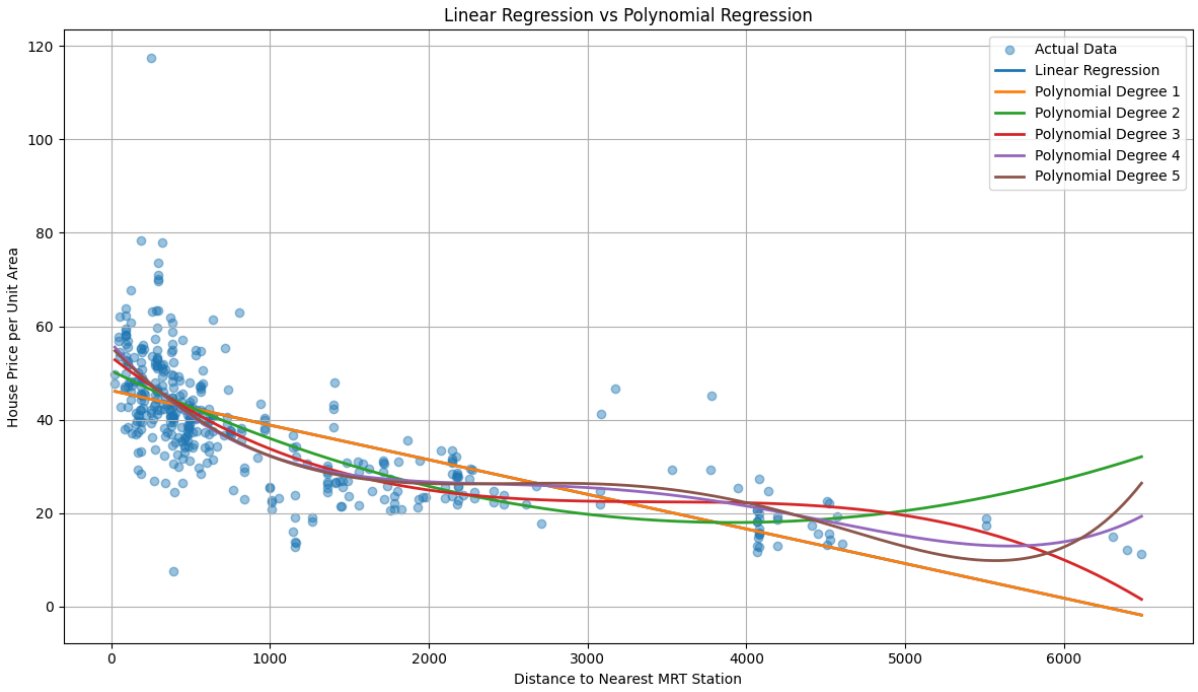
The original scatter plot displays the relationship between distance to the nearest MRT station and house price per unit area. Each point represents an observation in the dataset. The plot helps identify the general pattern of the data and assess whether a linear model may be sufficient.



Screenshot 7: Original dataset scatter plot

10.2 Comparison of Regression Curves

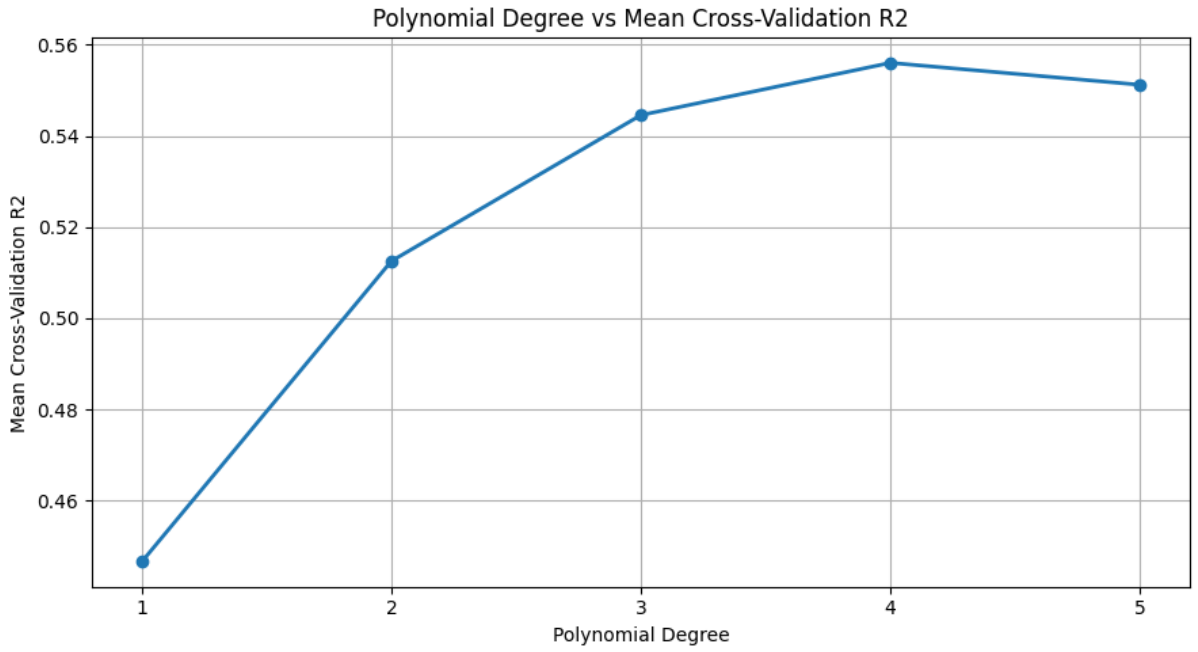
The regression curve visualization compares Linear Regression with Polynomial Regression models of Degrees 1 through 5. Linear Regression produces a straight line, while higher-degree polynomial models can produce curved relationships. The curves demonstrate how increasing the polynomial degree changes model flexibility. A more flexible curve is not necessarily better. It must improve predictive performance on unseen observations to justify its additional complexity.



Screenshot 8: All regression models visualized together

10.3 Cross-Validation Performance Graph

The cross-validation graph displays polynomial degree against mean cross-validation R^2. The graph provides a visual comparison of average model performance across the tested degrees. The degree with the highest mean cross-validation score is a candidate for model selection, but test performance, variability, and model complexity should also be considered.



Screenshot 9: Polynomial degree versus mean cross-validation R^2

11. Underfitting and Overfitting Analysis

Underfitting and overfitting were investigated by comparing training and testing performance.

11.1 Underfitting

In this experiment, the Linear Regression model achieved a training R^2 of **0.4317** and a test R^2 of **0.5390.** These results suggest that the baseline model may be underfitting, since even with a favorable test split it explains only **about 54%** of the variance in house price. The relatively low R^2 on both sets indicates that a straight line cannot fully capture the distance–price relationship. Polynomial Regression increases model flexibility by introducing higher powers of the input feature. Since higher-degree models improved both training and testing performance, this confirms that the baseline was too simple.

11.2 Overfitting

For Polynomial Regression Degree 4, the training R^2 was **0.5465**, while the test R^2 was **0.6443**. The difference between these scores was relatively small (a gap of **-0.0977**, meaning the model actually performed better on the test set). This suggests that the model does not show strong evidence of overfitting based on the selected train-test split. Note that the negative gap is a sampling artifact: with **only 83** test observations, the test set happens to contain easier-to-predict points than the training set.

11.3 Generalization

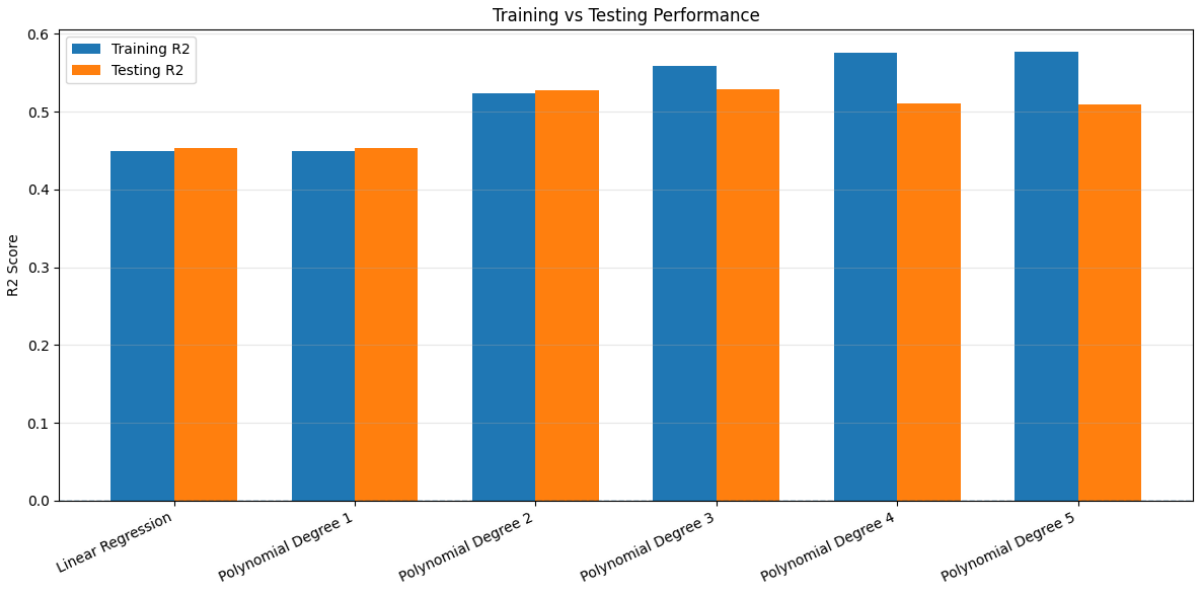
The model selected for further consideration was Degree **4**, because it achieved the strongest cross-validation performance (mean $R^2 = 0.5561$) and provided the best balance between test performance ($R^2 = 0.6443$, RMSE = 7.7250) and model complexity.

Underfitting and Overfitting Analysis

	Model	Training R2	Testing R2	R2 Gap	Training RMSE	Testing RMSE	Initial Diagnosis
0	Linear Regression	0.4496	0.4534	-0.0038	10.1646	9.5814	Possible Underfitting
1	Polynomial Degree 1	0.4496	0.4534	-0.0038	10.1646	9.5814	Possible Underfitting
2	Polynomial Degree 2	0.5233	0.5268	-0.0035	9.4596	8.9146	Reasonable but Weak Fit
3	Polynomial Degree 3	0.5588	0.5281	0.0307	9.1009	8.9029	Reasonable but Weak Fit
4	Polynomial Degree 4	0.5758	0.5105	0.0653	8.9240	9.0674	Reasonable but Weak Fit
5	Polynomial Degree 5	0.5763	0.5093	0.0670	8.9183	9.0787	Reasonable but Weak Fit

Screenshot 10: Underfitting and overfitting analysis table

"The diagnostic table shows that the baseline Linear Regression model produces a weak fit on both training and test data, while polynomial models of Degree 3 and higher achieve good generalization — high test R^2 with a small (and even negative) train–test gap. The negative gaps are a sampling artifact caused by the random 80/20 split: with only 83 test observations, the test set happened to contain points that follow the underlying curve more closely than some training points."



12. Prediction of a New Value

A new input value was supplied to the trained models to compare their predictions.The input feature was the distance to the nearest MRT station, and the predicted output was house price per unit area. The input distance used for the demonstration was **500.**
The models generated the following predictions:

Model	**Predicted House Price per Unit Area**
Linear Regression	41.83
Polynomial Degree 1	41.83
Polynomial Degree 2	42.12
Polynomial Degree 3	41.43
Polynomial Degree 4	40.10
Polynomial Degree 5	40.31

The predicted values differed because each model represented the relationship between distance and house price differently. The comparison demonstrates how polynomial degree can affect predictions for the same input. These values are model estimates, not guaranteed actual market prices. Predictions should be interpreted cautiously, particularly when the input falls outside the observed range of the training data.

```
Enter distance to the nearest MRT station: 500

Predicted House Prices
==================================================
Linear Regression: 41.83
Polynomial Degree 1: 41.83
Polynomial Degree 2: 42.12
Polynomial Degree 3: 41.43
Polynomial Degree 4: 40.10
Polynomial Degree 5: 40.31
```

Screenshot 11: Predictions from all trained models

13. Model Selection

The final model was selected by comparing test-set performance, cross-validation results, and model complexity.
The following considerations were used:

1. A higher test R^2 generally indicates that the model explains more variation in the test target.
2. Lower test RMSE and MAE indicate smaller prediction errors.
3. Higher mean cross-validation R^2 indicates better average validation performance under the selected cross-validation procedure.
4. A large gap between training and testing performance may indicate poor generalization.
5. A more complex model should be selected only when its performance justifies the additional complexity.

Based on the experimental results, Polynomial Regression Degree **4** was selected as the most promising model among those tested. It achieved a test R^2 of **0.6443**, a test RMSE of **7.7250**, and a mean cross-validation R^2 of **0.5561**. This selection reflects the observed results. It does not imply that the selected model will always perform best on other datasets or future real estate transactions.

14. Limitations

This experiment has several limitations.

- First, only one input feature was used. Real estate prices may depend on other factors, including house age, transaction date, number of convenience stores, and geographic location.
- Second, the models were evaluated using a particular train-test split. Different splits may produce different performance estimates.
- Third, the dataset contains 414 observations, so further validation would be useful before applying the results to practical real estate valuation.
- Fourth, Polynomial Regression can produce unreliable predictions outside the range of observed input values. Therefore, predictions made through extrapolation should be interpreted cautiously.

Finally, the target variable is house price per unit area, not necessarily the total market value of a property. The model should not be interpreted as directly predicting a property's total sale price.

15. Conclusion

This assignment implemented and compared Linear Regression with Polynomial Regression models of Degrees 1, 2, 3, 4, and 5 using the Real Estate Valuation dataset. The dataset was inspected before training, and no missing values or duplicate rows were detected. The distance to the nearest MRT station was selected as the input feature, while house price per unit area was selected as the target variable. The models were evaluated using R^2, Mean Squared Error, Root Mean Squared Error, and Mean Absolute Error. Five-fold cross-validation was also used to compare performance across different data splits.

The experimental results showed that Polynomial Regression outperformed Linear Regression for this dataset. While Linear Regression achieved a test R^2 of 0.5390 and RMSE of 8.7945, the best polynomial model (Degree 4) improved these to $R^2 = 0.6443$ and RMSE = 7.7250. Performance increased steadily from Degree 1 through Degree 4, but declined slightly at Degree 5, illustrating the bias–variance trade-off. Polynomial Regression Degree 4 was selected as the most promising model among those tested because it achieved the highest test R^2 (0.6443), the lowest test RMSE (7.7250), and the best mean cross-validation R^2 (0.5561), while maintaining a healthy training–test gap (−0.0977, indicating no overfitting on this split).

16. References

1. Lecture 8: Polynomial Regression, Machine Learning course lecture material.
2. Scikit-learn documentation for Linear Regression, PolynomialFeatures, Pipeline, model evaluation, and cross-validation.
3. Pandas documentation for dataset loading, inspection, and statistical analysis.
4. Matplotlib documentation for data visualization.
5. Real Estate Valuation dataset provided for this assignment.